

OCR Further Mathematics
Mechanics - Motion in a Circle
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question		Answer/Indicative content	Marks	Part marks and guidance		
1	a	$20 = 4u \Rightarrow u = 5$ Initial energy = $\frac{1}{2} \times 4 \times 5^2$ Energy at $\theta = \frac{1}{2} \times 4 \times v^2 + 4g \times 0.8(1 - \cos\theta)$ $2v^2 + 15.68 = 50 \Rightarrow v^2 = 17.16$ Radial: $a_r = \frac{v^2}{0.8} = \frac{17.16}{0.8}$ Tangential: $ma_t = -mg \sin \frac{\pi}{3}$ $a = \sqrt{\left(-\frac{\sqrt{3}g}{2}\right)^2 + \left(\frac{429}{20}\right)^2} = 23.067...$ so the magnitude of the acceleration is 23.1 m s^{-2} (3 sf)	B1 (AO1.1) B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO3.1b) M1 (AO3.1b) A1 (AO1.1) [7]	= 50 Attempt to derive total ME at general or specific angle Equating energies to derive a value for v^2 Correct form for centripetal acceleration and use of v^2 NII for tangential direction with weight resolved (– not necessary)	Assuming zero PE level at initial level of P $v = 4.142...$ $a_r = 21.45$ $a_t = -\frac{\sqrt{3}g}{2} = -8.4870...$	
	b	Radial: $T - 4g \cos\theta = \frac{4v^2}{0.8}$ $v^2 = 5^2 - 2g \times 0.8(1 - \cos\theta)$ $-7.84 \cos\theta = 9.32 + 15.68 \cos\theta$ $\therefore \cos\theta = -\frac{9.32}{23.52}$ $\therefore \theta = 113.3^\circ$ or 1.98 rads	M1 (AO2.1) M1 (AO2.1) A1 (AO3.2a) [3]	NII for radial direction. T could be set to 0. Correct form of a_r . v^2 in terms of from conservation of energy	$v^2 = 9.32 + 15.68 \cos\theta$	
		Total	10			



Question		Answer/Indicative content	Marks	Part marks and guidance		
2	a	KE of $P = \frac{1}{2}mv^2$ $\uparrow C \sin \theta = mg$ $\leftrightarrow C \cos \theta = ma$ $\frac{\cos \theta}{\sin \theta} = \frac{a}{g} = \frac{v^2}{rg}$ PE of P (exceeds that of Q by) $mgh = mg \frac{r}{\tan \theta} = mg \frac{r \cos \theta}{\sin \theta} = mg \frac{v^2}{g} = mv^2$ soi So that ME of P exceeds that of Q by $= mv^2 + \frac{1}{2}mv^2 = \frac{3}{2}mv^2 \text{ J}$	B1 (A01.2) M1 (A03.3) M1 (A03.3) M1 (A03.4) M1 (A03.4) A1 (A02.2a) [6]	Balancing forces in the vertical. C must be resolved NII in the horizontal using a resolved component of C Eliminating C (and m) between the two equations and using a correct form for a Using the relationship to find the (excess) PE of P in terms of m and v (and possibly g) only AG. Or total ME of $Q = 0$ but some justification of excess for PE at least must be seen in the solution	SSU – change C to R if a better reflection of candidate solutions In this solution, C is the normal contact force between P and the cone and θ is the semi-vertical angle of the cone May see $v^2 = gh$ here and used later h is the vertical height of P above Q Use R instead of C?	



Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		One of: – We have assumed that the radius of the circle which P moves in is the same as the radius of the cone at that level – Q is at V [neither of which is quite true if P and Q do not have a negligible radius]	B1 (AO3.5b) [1]	Also accept e.g. – CofM of P lies on the edge of the cone – CofM of Q lies at V	V is the vertex of the cone	
	c		Resistance to the motion of P should be included in the model.	B1 (AO3.5c) [1]	eg air resistance. Allow friction.		
			Total	8			



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Question		Answer/Indicative content	Marks	Part marks and guidance		
3	a	$\frac{1}{2} \times 2.8 \times 6^2 + 2.8 \times 9.8(0.75 - 0.75 \cos \theta) = \frac{1}{2} \times 2.8 v^2$ $50.4 + 20.58 - 20.58 \cos \theta = 1.4 v^2$ $v^2 = 50.7 - 14.7 \cos \theta$	M1 (AO3.3 1.1) M1 A1 (AO2.1) [3]	Attempt at GPE 0.75cos2.8g with correct theta and length Conservation of energy, allow sign errors AG An intermediate step must be shown	In this solution θ is the angle between OA and the upwards vertical through O	
	b	$T + 2.8 \times 9.8 \cos \theta = 2.8 \times \frac{v^2}{0.75}$ $200 + 2.8 \times 9.8 \cos \theta = 2.8 \times \frac{50.7 - 14.7 \cos \theta}{0.75}$ $\theta = 97.5^\circ \text{ or } 1.70 \text{ rad}$	M1 (AO3.3) A1 (AO1.1) M1 (AO2.2a) A1 (AO1.1) [4]	Using $F = ma$ in radial direction with tension, resolved weight and an expression for centripetal acceleration Using their expression for v^2 and breaking tension of string $\cos \theta = -\frac{134}{1029} = -0.1802...$	Allow sin instead of cos	
		Total	7			



Question		Answer/Indicative content	Marks	Part marks and guidance		
4	a	$\frac{1}{2}mv^2 + m\gamma r \cos \theta = m\gamma r \cos \alpha_{oe}$ $v^2 + 2\gamma r \cos \theta = \frac{3}{2}\gamma r$ $m\gamma \cos \theta (-C) = ma$ $a = \frac{v^2}{r}$ $\cos \theta = \frac{a}{\gamma} = \frac{v^2}{\gamma r}$ $v^2 = \frac{1}{2}\gamma r$ $r \cos \theta = \left(\sqrt{\frac{1}{2}\gamma r \sin \theta} \right) t + \frac{1}{2}\gamma t^2$ $t^2 + \sqrt{\frac{3r}{2\gamma}}t - \frac{r}{\gamma} = 0$ $t = \frac{1}{4}\sqrt{\frac{r}{\gamma}}(\sqrt{22} - \sqrt{6})$ $OF = r \sin \theta + \sqrt{\frac{1}{2}\gamma r \cos \theta} \times \frac{1}{4}\sqrt{\frac{r}{\gamma}}(\sqrt{22} - \sqrt{6})$ $OF = \frac{r}{8}(\sqrt{11} + 3\sqrt{3})_{oe}$	M1(A03.3) A1(A01.1) B1(A01.1) B1(A01.1) M1(A02.2a) A1(A02.2a) M1(A03.4) *M1(A01.1) Dep *M1(A01.1) Dep *M1(A03.4) A1(A01.1) [11]	Conservation of energy Nil for P at point where it is about to lose contact with surface. soi Use previous two results to relate v and $\cos \theta$ for v or v^2 or $\cos \theta = \frac{1}{2}$ Use of $s = ut + \frac{1}{2}at^2$ using their vertical component of v as u where v has come from consideration of theta Reduction to 3 term quadratic with numerical values for trig ratios Solve for t (BC)	θ is the angle between OP and the upward vertical Could see contact force, C, later set to 0 Or use of trajectory eqn: $y = x \tan \theta + \frac{\gamma x^2}{2v^2 \cos^2 \theta}$ with $y = r$ $\cos \theta$ and $v^2 = \frac{1}{2}\gamma r$ $8x^2 + 2\sqrt{3}rx - r^2 = 0$ $x = \frac{\sqrt{11} - \sqrt{3}}{8}r$ $r \sin \theta + \frac{\sqrt{11} - \sqrt{3}}{8}r$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					Find OF	
					<u>Examiner's Comments</u> Candidates found this question difficult but most candidates were given some marks. Higher ability candidates drew good diagrams and considered conservation of energy between the start point and the point where the particle leaves the surface. Newton's Second Law was then used to find $\cos \theta$ and an expression for v . These candidates were given 6 marks for doing this successfully. Not many candidates were given any further marks for considering vertical motion to find an expression for t and then considering horizontal motion (including adding on $r \sin \theta$). In this second part of the question candidates who took time to organise their projectile equations generally made better progress. This was a challenging question where the best candidates were able to produce a detailed and well-structured solution. Some candidates set the start point at the top of the hemisphere and that the particle left the hemisphere at α which made it very difficult for them to be given any marks. Incorrect signs were used in the conservation of energy equation (mainly from	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					candidates having difficulties understanding their zero level of potential energy) and other candidates didn't introduce another angle (i.e. θ) where the particle left the surface.	
	b		Unchanged since OF does not depend on γ	E1(A03.5a) [1]	 <u>Examiner's Comments</u> If candidates had worked out part (a) correctly they were able to argue that OF didn't depend on gravity so it would remain unchanged.	
			Total	12		



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Question		Answer/Indicative content	Marks	Part marks and guidance		
5	a	$\cos \theta = \frac{1}{2}$ or $\sin \theta = \frac{\sqrt{3}}{2}$ $v \sin \theta = (1+m)V$ $V = \frac{\sqrt{3}v}{2(1+m)}$ $I = mV$ or $I = \pm(V - v \sin \theta)$ $V = \frac{v \times \frac{1}{2} \sqrt{3}}{1+m} \Rightarrow I = \frac{\sqrt{3}mv}{2(1+m)}$	B1(A03.1b) M1(A03.1b) A1(A03.1b) A1(A02.2a) [4]	Must be clear where it comes from ie. Shown in diagram or clear use of distances. May be other way round Conservation of momentum in direction of string Working must be seen AG Justification of using Impulse = change in momentum	θ is the angle between the string and the direction of AB when the string initially becomes taut	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					to find an equation for the final velocity in terms of m and v. Most candidates were unable to show logical reasoning and seemed to use the answer given in the question to form their answer. For the final mark examiners had to be certain that the candidate had considered change of momentum on either particle. Just multiplying their expression for velocity by 'm' wasn't enough to demonstrate this. A significant amount of candidates were unable to access this question because they were trying to model the problem in 1 dimension.  AFL When answering a question (particularly 'show that') encourage candidates to write what they are using to set up the equation and in what direction to communicate clearly what the intention is. For instance: 'CoM parallel to the string' or 'Impulse on A'	



Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	$KE = \frac{1}{2}(v^2 + (v \cos \theta)^2)_{\text{soi}}$ $KE = \frac{1}{2} \left(\left(\frac{\sqrt{3}v}{2(1+m)} \right)^2 + \left(v \times \frac{1}{2} \right)^2 \right)$ $KE = \frac{v^2(4 + 2m + m^2)}{8(1+m)^2}$	M1(A03.1b) M1(A01.1) A1(A01.1) [3]	Transverse component unchanged and using their longitudinal component Substituting in for V and $\cos \theta$ (could just be in speed equation) Or any equivalent single algebraic fraction	Condone consideration of speed squared or inclusion of m in KE	

Examiner's Comments

This was the most difficult question on the paper for candidates to access. To be given the first 2 marks candidates needed to identify the 2 components of B's final speed and square each of them. Out of the minority of candidates who managed this, the main error was to forget that $m = 1$ in the kinetic energy formula leading to an extra m in their final answer.



Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$V = \frac{\sqrt{3}v}{2(1+m)} \rightarrow 0 \text{ as } m \rightarrow \infty$ <i>B</i> moves (approximately) in a circle around <i>A</i>	B1(A03.2a) B1(A02.2b) [2]	If <i>m</i> is very large then <i>A</i> is approximately stationary or <i>B</i> has only its transverse velocity of $\frac{1}{2}v$ after the string becomes taut		
					<u>Examiner's Comments</u> Some candidates omitted this question after struggling with the first 2 parts; however this question was accessible if candidates had modelled the situation. Essentially if <i>m</i> is very large, then <i>A</i> will be stationary as <i>B</i> will be unable to move it. As <i>A</i> is a fixed point then <i>B</i> will travel in circular motion around it. Some candidates were able to infer circular motion but didn't refer to <i>A</i> being stationary. The most common incorrect answers were that <i>B</i> would stop or performing simple harmonic motion.		
			Total	9			

$T \cos \theta + R \sin \theta = mg$	*M1(A03.3)	m, g, r and / or T may appear as	Must have
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Question			Answer/Indicative content	Marks	Part marks and guidance	
					Us $a = \frac{v^2}{r}$ to e eliminate a and introduce v Rearrange to make v^2 the subject or solve for v <u>Examiner's Comments</u> The majority of candidates were able to get some marks on this question, although few received full marks. Pretty much all the candidates were able to find the trig ratios and use $a = \frac{v^2}{r}$ for 2 marks. Fewer candidates used the correct value for the radius. Some candidates were able to set up equations to solve. Common mistakes were a lack of understanding of resolving components and a misunderstanding of the direction of acceleration. Some candidates discounted R or did not resolve it. Candidates who didn't draw a good force diagram significantly disadvantaged themselves here.	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance		
7		a	$\frac{1}{2}m \times 3.5^2 = \frac{1}{2}mv^2 + mg \times 0.8(1 - \cos \frac{1}{3}\pi)$ $v^2 = 3.5^2 - 0.8g = 4.41$ $T - 1.2g \cos \frac{1}{3}\pi = \frac{1.2v^2}{0.8}$ So tension when string breaks is awrt 12.5N	M1 (AO 3.1b) A1 (AO 1.1) M1 (AO 3.1b) A1 (AO 3.2a) [4]	Conservati on of energy (could be general angle for RHS) Or $v = 2.1$ Resolving weight and use of NII with correct centripetal acceleratio n		
		b	$v_y = 2.1 \sin \frac{1}{3}\pi$ upwards P starts freefall $1.5 + 0.8(1 - \cos \frac{1}{3}\pi)$ above plane $-(1.5 + 0.8(1 - \cos \frac{1}{3}\pi)) = (2.1 \sin \frac{1}{3}\pi)t - \frac{1}{2}gt^2$ $-1.9 = (1.05\sqrt{3})t - 4.9t^2$ or $4.9t^2 - 1.8186t - 1.9 = 0$ $t = \text{awrt } 0.835$ Horizontal distance in freefall is $2.1 \cos \frac{1}{3}\pi \times 0.835 \dots$ $0.8 \sin \frac{1}{3}\pi + 2.1 \cos \frac{1}{3}\pi \times 0.835 \dots = \text{awrt } 1.57 \text{ m}$	M1 (AO 3.1b) B1 (AO 3.1b) M1 (AO 1.1) A1 (AO 1.1) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.2a)	Direction can be implied by eg use of - 9.8 (= 1.9) Setting up 3 term quadratic equation in t with adjustment to ± 1.5 as displaceme nt Correct equation soi BC 0.877...	$\frac{\sqrt{331} + 3\sqrt{3}}{28}$ $\frac{3\sqrt{331} + 9\sqrt{3}}{80}$	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			Alternative solution Projection velocity is 2.1ms^{-1} at $\frac{1}{3}\pi$ above horizontal P starts freefall $1.5 + 0.8(1 - \cos\frac{1}{3}\pi)$ above plane Equation of trajectory: $y = x \tan \frac{1}{3}\pi - \frac{gx^2}{2 \times 2.1^2 \cos^2(\frac{1}{3}\pi)}$ $-1.9 = x\sqrt{3} - \frac{40}{9}x^2$ or $4.444x^2 - 1.732x - 1.9 = 0$ $x = \text{awrt } 0.877$ $0.8 \sin \frac{1}{3}\pi + 0.877... = \text{awrt } 1.57\text{m}$	B1ft B1 M1 A1ft M1 A1 A1 [7]	horizontal distance while attached plus horizontal distance in freefall. soi; ft their 2.1 (= 1.9) Or suitably adjusted if not using P's position when string breaks as the origin Correct equation (with their 2.1) soi oe: formulation of explicit quadratic equation in x BC	$\frac{3\sqrt{331} + 9\sqrt{3}}{80}$	
			Total	11			



Question			Answer/Indicative content	Marks	Part marks and guidance		
8		a	$\frac{1}{2}mu^2 = mgh(1 - \cos \theta)$ $mu = 2mv_B$ $e = \frac{v_B}{u}$ $v_B = \frac{1}{2}u$ and $e = \frac{1}{2}$ $2m \times \frac{1}{2}u = mV_A + 2mV_B$ $\frac{1}{2} = \frac{V_A - V_B}{\frac{1}{2}u}$ $V_A = \frac{1}{2}u$ $\frac{1}{2}m\left(\frac{1}{2}u\right)^2 = mgh(1 - \cos \phi)$ $\frac{1}{8}mu^2 = \frac{1}{4}mgh(1 - \cos \theta) = mgh(1 - \cos \phi)$ $\Rightarrow \cos \phi = \frac{3 + \cos \theta}{4}$	M1 (AO 3.3) M1 (AO 3.4) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1) M1 (AO 3.4) A1 (AO 2.1) [9]	Conservation of energy for A Conservation of momentum Restitution Conservation of momentum for 2nd collision with unchanged v_B Restitution for 2nd collision Conservation of energy for A AG so elimination of $\frac{1}{2}mu^2$ or must be clearly shown	v_B is velocity of B after first collision	

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$V_B = \frac{1}{4}u$ $2m \times \frac{1}{4}u + m \times \left(-\frac{1}{2}u\right) = mv_A + 2mv_B$ $\frac{1}{2} = \frac{v_A - v_B}{\frac{1}{4}u - \left(-\frac{1}{2}u\right)}$ $v_B = -\frac{1}{8}u \text{ and } v_A = \frac{1}{4}u$ $\text{KE loss} = \frac{\frac{1}{2}m \times \left(\frac{1}{4}u\right)^2 + \frac{1}{2} \times 2m \times \left(-\frac{1}{8}u\right)^2}{\frac{1}{2}mu^2} = \frac{3}{32} = 9.4\%$ So percentage of KE remaining is 90.6%	B1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1) A1 (AO 1.1) [5]	Conservation of momentum for 3rd collision Restitution for 3rd collision	Or with signs reversed, if different 'positive' direction chosen	
		c	During and between the 1st and 2nd collisions there is no external horizontal force acting on A and while there is an external force acting on B its velocity and hence momentum is the same just before the 2nd collision as it is just after the 1st collision	E1 (AO 2.4) [1]	Identifying the unchanged velocity for B in the two collisions		
		d	Between the 2nd and 3rd collisions a component of the tension in the string OA is horizontal and so the momentum of the system is not conserved	E1 (AO 2.4) [1]	Identifying the relevant external force acting		
			Total	16			



Question			Answer/Indicative content	Marks	Part marks and guidance		
9		a	$T = 365 \times 24 \times 60 \times 60 \text{ s}$ and $R = 1.5 \times 10^{11} \text{ m}$ $\omega = \frac{2\pi}{T}$ or $v = \frac{2\pi R}{T}$ $F = mR\omega^2$ or $F = \frac{mv^2}{R}$ $F = 5.97 \times 10^{24} \times 1.5 \times 10^{11} \times (1.99 \times 10^{-7})^2$ oe 3.5×10^{22} or $3.6 \times 10^{22} \text{ N}$	B1 (AO 1.1a) M1 (AO 1.1) M1 (AO 1.1) M1 (AO 1.1) A1 (Ao 1.1) [5]	Time converted to s and radius to m Used (units of T could be wrong) Used (units could be wrong) FT their values, <i>provided that</i> there was a proper attempt at conversion of units cao	$(T = 31536000 \text{ s})$ $(\omega = 1.99 \times 10^{-7} \text{ s}^{-1})$ $(v = 2.99 \times 10^4 \text{ ms}^{-1})$	
		b	[The force is directed] towards the sun	B1 (Ao 1.1) [1]			
			Total	6			



Question			Answer/Indicative content	Marks	Part marks and guidance		
10		a	$\frac{1}{2} \times 0.4 v_B^2 = \frac{1}{2} \times 0.4 \times 0.8^2 + 0.4g \times 1.4(\cos 30^\circ - \cos 60^\circ)$ $v_B^2 = 10.68...$ $R - 0.4g \cos 30^\circ = 0.4 \times \frac{v_B^2}{1.4}$ Magnitude of contact force at B is 6.45N	M1(A03.3) A1(A01.1) M1(A03.3) A1ft(A01.1) A1(A01.1)	Conservati on of energy from A to B soi Use of Newton II radially, 3 terms Correct equation, with their v_B		
				[5]	cao		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Vertical velocity component at B is $v_B \sin 30^\circ$ Height of B above plane is $2.5 + 1.4(1 - \cos 30^\circ)$ Time of flight given by $-2.687 = 1.634t - \frac{1}{2}gt^2$ $t = 0.9259...$ Horizontal distance covered in flight is $(v_B \cos 30^\circ)t$ $FH = 1.4\sin 30^\circ + 2.6209... = 3.32\text{m}$ Alternative solution Height of B above plane is $2.5 + 1.4(1 - \cos 30^\circ)$ Projection velocity is v_B at 30° above horizontal Equation of trajectory is $0.61152x^2 - 0.57735x - 2.68756 = 0$ $x = 2.62...$ $FH = 1.4\sin 30^\circ + 2.6209... = 3.32\text{m}$	B1ft(AO3.1b) B1(AO1.1) M1(AO3.4) A1(AO1.1) A1(AO1.1) M1(AO3.4) A1(AO3.2a) B1 B1ft M1 A1 M1 A1 A1 [7]	$s = ut + \frac{1}{2}at^2$ U_vse_erti_ocally Correct equation BC Using their values of v_B and t oe, if origin other than B is used Correct equation, with their v_B Formation of explicit quadratic in x BC	(1.634...) (2.687...) (2.6209...) (2.687...)	
			Total	12			



Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	$T \sin \theta = 3.5 \times 3^2 \times r$ $r = L \sin \theta$ $T \sin \theta = 3.5 \times 3^2 \times L \sin \theta$ oe $T = 31.5L$	M1(A03.3) B1(A01.1) M1(A01.1) A1(A01.1) [4]	Resolving tension and using NII Eliminating r AG working must be clear	may use $T \cos \phi$ instead or $r = L \cos \phi$	
		b	$T = \frac{75(L - 0.8)}{0.8}$	B1(A03.3) [1]	Correct use of Hooke's law	$(T = 93.75L - 75)$	
		c	$31.5L = 93.75(L - 0.8)$ $T = 38.0$ $L = 1.20$	M1(A01.1a) A1(A01.1) A1(A01.1) [3]	Solution of simultaneous equations		
		d	$T \cos \theta = mg$ 25.3°	M1(A03.4) A1(A01.1) [2]	Resolve vertically		
			Total	10			



Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	$\sin \theta = \frac{4}{5}$ or $\cos \theta = \frac{3}{5}$ or $\tan \theta = \frac{4}{3}$ or $\theta = 53.1^\circ$ $R \sin \theta = 0.4g$ $R = \frac{1}{2}g$	B1 M1 A1 [3]	θ is the angle above the horizontal, may be awarded in (ii) or (iii) unless different value already used in (i) Resolving vertically with mass substituted, angle need not be substituted for M1; allow use of their angle including 45° Examiner's Comments Candidates generally knew what to do here, and the question was usually done well; a few resolved the weight instead of the force, some could not find the angle, and some gave the answer as 4.9 when the request was to find the magnitude of the force in terms of g.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$r = 3$ $R \cos \theta = 0.4(3)\omega^2 \text{ or } \frac{0.4v^2}{3}$ $[\tan \theta = \frac{4}{3} \Rightarrow \omega^2 = \frac{g}{4} \quad \tan \theta = \frac{4}{3} \Rightarrow \omega^2 = \frac{g}{4} \text{ or } v^2 = \frac{9g}{4}]$ $\text{Time} = \frac{2\pi}{\sqrt{\frac{g}{4}}} \left(\text{or } \frac{2\pi(3)}{\sqrt{\frac{9g}{4}}} \right) = \frac{4\pi}{\sqrt{g}}$	B1 *M1 dep*M1 A1 [4]	May be awarded in (iii) unless different value already used in (ii) N2L with $cv(r)$ Substituting for R and θ Use of $\frac{2\pi}{\omega}$ or $\frac{2\pi r}{v}$ to obtain AG Correctly shown Examiner's Comments Generally well done, with the first 3 marks being earned quickly. The final mark sometimes proved too difficult, mainly for those who were not working in terms of g . The value of r was not always found and when it was, it was not always 3. Formulae such as $a = r\omega^2$ and $R \cos \theta = mr\omega^2$ often written down but candidates then made no attempt to substitute values. A number of candidates produced unconvincing final steps or used statements to 'show' a decimal answer was the same as the AG. Some candidates did realise at the end that earlier working needed to be converted back to g form.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\sin \alpha = \frac{1}{\sqrt{10}}$ or $\cos \alpha = \frac{3}{\sqrt{10}}$ or $\tan \alpha = 1/3$ or $\alpha = 18.4^\circ$ $R \sin \theta = T \sin \alpha + 0.4g$ $\frac{4}{5}R = \sqrt{40} \frac{1}{\sqrt{10}} + 0.4g$ $R \cos \theta + T \cos \alpha = \frac{0.4v^2}{3}$ or $0.4(3)\omega^2$ $v = 8.85 \text{ m s}^{-1}$	B1 M1 A1 M1 A1 A1 [6]	α is the angle below the horizontal Resolving vertically, 3 terms, use of different angles $R = 7.4$ N2L horizontally, 3 terms, use of different angles, $a = v^2/r$ or $r\omega^2$ $cv(r)$, do not need a value of R here 8.8487287... note that $\omega = 2.94957...$ Examiner's Comments This proved a difficult request for a significant number of candidates. The problems encountered were basic understanding that this is a new situation compared to earlier and a new normal contact force was required to be calculated. Also many did not realise that the angle that the string makes with the horizontal (or vertical) was required, many using the earlier angle calculated or, in a minority of cases, using 45° .	
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$I = \frac{1}{3}(2M)(2a)^2 + 2M(ka)^2$ $T = 2\pi \sqrt{\frac{8Ma^2 + 6Mk^2a^2}{3(2Mg)(ka)}}$ $T = 2\pi \sqrt{\frac{a}{3g} \left(\frac{4 + 3k^2}{k} \right)}$	B1 B1 M1 A1 [4]	B1 for each term SC B1: for incorrect masses in both expression $T = 2\pi \sqrt{\frac{I}{mgh}}$ with correct mass AG www	Examiner's Comments Part (i) was answered extremely well with the majority of candidates correctly applying the standard formula for the period of small oscillations of a compound pendulum. The main issue came in the calculation of the moment of inertia about the horizontal axis when a number of candidates $\frac{2M(2a)^2}{3} \text{ to } \frac{1}{3}(2M)(2a)^2$ Furthermore, a number of candidates incorrectly added values for the mass and length of the rod.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{d}{dk} \left(\frac{4+3k^2}{3k} \right) = 0$ $\frac{3k(6k) - (4+3k^2)(3)}{(3k)^2} = 0 \Rightarrow k^2 = \dots$ $k^2 = \frac{4}{3}$	M1 M1 A1 [3]	Attempt to differentiate $f(k)$ and set equal to zero Correct application of quotient rule (or other correct method) and get to $k^2 = \dots$ or $k = \dots$ (if k^2 not stated first) Cao (oe) Examiner's Comments In part (ii) the vast majority understood the need to differentiate, with respect to k, the expression $\frac{4+3k^2}{k}$ and to then set this answer equal to zero to find the value of k^2 for which the period of oscillations would be least. A number of candidates made this part significantly more demanding for themselves by attempting to differentiate the whole expression for T. Some candidates did not read this part carefully and gave the value for k without ever stating the value for k^2.	Or differentiate s wrt k eg $\frac{d}{dk} \left(\frac{4}{3k} + k \right) = 0$
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance		
14		i	$3mga \cos \frac{\pi}{6}$ and $2mga \cos \frac{\pi}{6}$ $3mga \cos \left(\frac{\pi}{6} + \theta \right) + 2mga \cos \left(\frac{\pi}{6} - \theta \right) + \frac{1}{2} 3mv^2 + \frac{1}{2} 2mv^2$ $v^2 = \frac{2}{5} ag \left(5 \cos \frac{\pi}{6} - 3 \cos \left(\frac{\pi}{6} + \theta \right) - 2 \cos \left(\frac{\pi}{6} - \theta \right) \right)$	B1 M1 A1 A1 [4]	Initial PE Final PE + KE AG Equating and correct manipulation	If O is zero level for PE For M1 at least 1 KE and 1 PE term; allow m used for 2m/3m; wrong signs; missing g	
					Examiner's Comments There were two main approaches to this part. Some candidates equated the total potential energy of A and B in the initial position with their total energy when angle AOT is $\left(\frac{1}{6} \pi + \theta \right)$ radians; other candidates considered the increase in potential energy of B, the decrease in potential energy of A and the increase in kinetic energy of both particles. Either approach, explained clearly, could gain full marks. However, some candidates did not show enough working to convince examiners that they deserved full marks.		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v^2 = \frac{2}{5}ag \left(5 \cos \frac{\pi}{6} - 3 \cos \frac{\pi}{3} - 2 \cos 0 \right)$ $3mg \cos \frac{\pi}{3} - R = 3m \frac{v^2}{a}$ $R = 3mg \cos \frac{\pi}{3} - 3m \frac{2g}{5} \left(5 \cos \frac{\pi}{6} - \frac{7}{2} \right)$ $R = mg \left(5.7 - 6 \cos \frac{\pi}{6} \right) \text{ oe}$	B1 M1 A1 A1 [4]	$v^2 = \frac{1}{5}ag(5\sqrt{3} - 7)$ OR $3mg \cos \left(\frac{\pi}{6} + \theta \right) - R = 3m \frac{v^2}{a}$ $R = 3mg \cos \left(\frac{\pi}{6} + \theta \right) - 3m \frac{2}{5}g \left(5 \cos \frac{\pi}{6} - 3 \cos \left(\frac{\pi}{6} + \theta \right) - 2 \cos \left(\frac{\pi}{6} - \theta \right) \right)$ $R = mg(6.6 \cos \left(\frac{\pi}{6} + \theta \right) + 2.4 \cos \left(\frac{\pi}{6} - \theta \right) - 6 \cos \frac{\pi}{6})$ Answer must be simplified 4.94m loses last mark Accept mg or mg(5.7 - 3√3) oe Examiner's Comments Most candidates realised that they needed to use Newton's Second Law for particle A when angle TOA is $\frac{1}{3}\pi$, but only a minority of candidates arrived at the correct answer. The most common error was to have the wrong mass in one or more of the terms in the Newton's Second Law equation; generally to use m but $2m$ and $5m$ were also seen. The other common error was not to be able to realise that when particle B passes through point T the angle θ is 30°.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance		
15		a	$35g(5) - \frac{1}{2}(35)v^2 = \dots$ $\dots 35g(7 - 4\cos\theta)$ $v^2 = 4g(2\cos\theta - 1)$	M1(A03.3) A1(A01.1) A1(A01.1) A1(A01.1) E1(A02.1) [5]	Attempt at conservation of mechanical energy PE term on lhs KE term on rhs www; AG must show an intermediate step		
		b	When $\theta = 0^\circ$, $v_H = 2\sqrt{g}$ $v_v^2 = 2g(3)$ $v = 9.90$ so speed is 9.90 m s^{-1} $\theta = 50.8^\circ$ below the horizontal	B1(A03.3) B1(A03.3) M1(A03.4) A1(A01.1) A1(A01.1) [5]	Use of either $v^2 = v_H^2 + v_v^2$ or $\tan\theta = \frac{v_v}{v_H}$	6.260990... 9.8994949 ... 50.768479 ...	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	Tension when $\theta = 60^\circ$ is 17.5g N $T - 35g \cos \theta = \frac{35}{4}(4g(2 \cos \theta - 1))$ $\theta = 33.6^\circ$	B1(A01.1) *M1(A03.1b) A1(A01.1) dep*M1(A01.1a) A1(A01.1) [5]	May appear anywhere in answer Attempt to use N2L radially with $a = \frac{v^2}{r}$ substituting 3×their(T) into their N2L attempt and solving for θ BC	N2L is Newton's second law 33.55730...	
			Total	15			



Question			Answer/Indicative content	Marks	Part marks and guidance	
16			$I = \frac{1}{3}(5)(0.1^2 + 0.1^2) \left(= \frac{1}{30} \right)$ $\alpha = \frac{0.06}{\left(\frac{1}{30} \right)} (= 1.8)$ $\omega^2 = 8^2 + 2(1.8)(155)$ $\omega = 24.9 \text{ (3sf)}$	B1 M1 M1 A1	M1 for using $C = I\alpha$ with their I Using $\omega^2 = \omega_0^2 + 2\alpha\theta$ with their α accept $\sqrt{622}$	Work-energy principle (M1 A1ft) $\frac{1}{2}\left(\frac{1}{30}\right)\omega^2 - \frac{1}{2}\left(\frac{1}{30}\right)8^2 = 155(0.06)$ 24.939927... Examiner's Comments The majority of candidates correctly calculated the moment of inertia of the square lamina about an axis passing through its centre and then either correctly applied $C = I\alpha$ together with $\omega^2 = \omega_0^2 + 2\alpha\theta$ or correctly applied the work-energy principle to find ω. The most common errors were in the calculation of the moment of inertia of the lamina with a number of candidates using 0.2 rather than 0.1 in the formula $\frac{1}{3}m(a^2 + b^2)$ or incorrectly finding the moment of inertia about a vertex rather than the centre of the square lamina.
			Total	4		



Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	4 revolutions $\Rightarrow \theta = 8\pi$	B1		
		i	$12.4^2 = 3^2 + 2\alpha(8\pi)$	M1	Using $\omega^2 = \omega_0^2 + 2\alpha\theta$ with cv($\theta(\neq 4)$)	
		i	$\alpha = 2.88 \text{ rad s}^{-2} \text{ (3 s f)}$	A1	Examiner's Comments Nearly all candidates found this first question straightforward and the vast majority found the correct angular acceleration of the turntable in this part. The most common error was stating the angular displacement as 4 rather than 8π .	
		ii	$8\pi = \frac{1}{2}(3 + 12.4)t$	M1	Using $\theta = \frac{1}{2}(\omega_0 + \omega)t$ with cv(θ) (allow any θ)	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$t = 3.26 \text{ s (3 s f)}$	A1	Examiner's Comments Nearly all candidates found this first question straightforward and the vast majority found the time taken for the turntable to increase its angular speed in this part. The most common error was stating the angular displacement as 4 rather than 8π. It is worth pointing out that candidates are advised, where possible, to use methods that do not rely on previous obtained results. For example, in part (ii) a number of candidates used the formula $\omega = \omega + \alpha t$ with an incorrect value of α to calculate t. Candidates could have bypassed this incorrect value altogether by using the formula $\theta = \frac{1}{2}(\omega_0 + \omega)t$ which is independent of the angular acceleration.	
Total				5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	MI square centre $I_c = \frac{1}{3}m(a^2 + a^2)$	B1	May be implied by later working Examiner's Comments Most candidates correctly found the moment of inertia of the square lamina by applying either the perpendicular or parallel axis rule.	$B1 I_x = \frac{4}{3}ma^2$
		i	$I_A = \frac{1}{3}m(2a^2) + m(a\sqrt{2})^2 = \frac{8}{3}ma^2$	M1A1	M1 for applying parallel axes rule	M1 applying perpendicular axes rule ($I_x = I_y$)
		ii	$\frac{1}{2}I\omega^2 = mga\sqrt{2}(1 - \cos\theta)$	M1	Equation involving KE and PE Examiner's Comments Almost all candidates then went on to derive the correct form for the angular speed given.	
		ii	$\frac{1}{2}\left(\frac{8}{3}ma^2\right)\omega^2 = mga\sqrt{2}(1 - \cos\theta)$	A1 A1	A1 for KE term, A1 for PE term	
		ii	$a\omega^2 = \frac{3}{4}g\sqrt{2}(1 - \cos\theta)$	A1	AG Correctly obtained	
		iii	$2a\omega \frac{d\omega}{dt} = \frac{3g\sqrt{2}}{4} \sin\theta \frac{d\theta}{dt}$	M1	Differentiating ω with respect to t Examiner's Comments Many candidates went on to form the correct equations of motion for X and Y. The most common mistakes included sign errors, omission of the mass and either omission, or an incorrect, radius. However, the radial and transverse accelerations were almost always correct.	Or M1 for applying $C = I\alpha$ with their I from (i)



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\alpha = \frac{3g\sqrt{2}}{8a} \sin \theta$	A1		A1 for
		iii	$mg \cos \theta - X = ma\sqrt{2}\omega^2$	M1	For radial acceleration $r\omega^2$	
		iii	$X = \frac{1}{2}mg(5 \cos \theta - 3)$	A1		
		iii	$mg \sin \theta - Y = ma\sqrt{2}\alpha$	M1	For transverse acceleration $r\alpha$	
		iii	$Y = \frac{1}{4}mg \sin \theta$	A1		
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
19		i	$I_{rod} = \frac{4}{3}(4m)(2a)^2 \left(= \frac{64}{3}ma^2 \right)$	B1	Examiner's Comments	
		i	$I_{shell} = \frac{2}{3}ma^2 + m(5a)^2$	M1A1	The moment of inertia of the pendulum was almost always found correctly. Attempt at MI of shell about the axis A using the parallel axes rule	
		i	$I = \frac{66}{3}ma^2 + 25ma^2 = 47ma^2$	A1	AG Correctly shown	
		ii	Angular momentum of particle before impact $= 3a(m\sqrt{kg a})$	B1	Examiner's Comments	
		ii	Angular momentum after impact = $(47ma^2 + m(3a)^2)\omega$	B1	Many candidates did not realise that the correct approach was to apply conservation of angular momentum and a significant number attempted to use an energy approach. Those that did realise that angular momentum was required usually went on to derive the given result correctly.	
		ii	By conservation of angular momentum	M1		
		ii	$3ma\sqrt{kg a} = 56ma^2\omega$ $\omega = \frac{3}{56}\sqrt{\frac{kg}{a}}$	A1	AG Correctly shown	
		iii	By conservation of energy			



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$-16mga + \frac{1}{2}(56ma^2)\left(\frac{3}{56}\sqrt{\frac{kg}{a}}\right)^2 = 0 - 16mga \cos \theta$	B1* B1	B1* for kinetic energy, B1 for potential energy Examiner's Comments Candidates found (iii) and (iv) the most demanding parts on the whole paper and only a few succeeded in getting both parts correct. While the majority of candidates realised that the correct approach was to apply the conservation of energy there were numerous errors seen. These included not calculating the correct position for the centre of mass of the pendulum, using m rather than $6m$ for the mass of the pendulum, and using the mass rather than the moment of inertia in the calculation of the kinetic energy.	
		iii	$16 \cos \theta = 16 - 112 \left(\frac{3}{56}\right)^2$	M1 dep*	Conservation of energy – must be using correct mass and V must be of the correct form	
		iii	$\theta = 0.201$	A1	0.2007830...	Degrees: 11.504019...
		iv	$16mga = \frac{1}{2}(56ma^2)\omega^2$	M1	KE loss = PE gain	Must be using correct mass of $6m$
		iv	$k = 199$	A1	Cao (accept $k = 200$)	$k = \frac{1792}{9} = 199.111...$
			Total	14		



Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	$76 = 42 + \alpha \times 8$	M1	Using $\omega_1 = \omega_0 + \alpha t$	
		i	Angular acceleration is 4.25 rads ⁻²	A1		
		ii		M1	Using $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$	
		ii	$810 = 42t + 2.125t^2$ $t = \frac{-42 \pm \sqrt{42^2 + 4 \times 2.125 \times 810}}{2 \times 2.125}$	A1	FT Quadratic equation for t	
		ii	OR $\omega_1^2 = 42^2 + 2 \times 4.25 \times 810$ $\omega_1 = 93$ $93 = 42 + 4.25t$		M1 $\omega_1^2 = \omega_0^2 + 2\alpha\theta$ and $\omega_1 = \omega_0 + \alpha t$ A1 FT Equation for t	Or equivalent
		ii	Time is 12 s	A1	<u>Examiner's Comments</u> This question, on constant angular acceleration, was almost always answered correctly.	
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i		B1	For	
					$I_G = \frac{1}{3}m[(3a)^2 + (4a)^2]$	
		i	$I = \frac{1}{3}m[(3a)^2 + (4a)^2] + m(5a)^2$	M1	For $I_G + m(AG)^2$	
		i	OR	B1	For $I_{AD} = \frac{4}{3}m(3a)^2, I_{AB} = \frac{4}{3}m(4a)^2$	
		i	$I = \frac{4}{3}m(3a)^2 + \frac{4}{3}m(4a)^2$	M1	For $I_{AD} + I_{AB}$	
		i	$I = \frac{100}{3}ma^2$	A1	<u>Examiner's Comments</u> In part (i) the moment of inertia of the lamina was almost always found correctly.	
		ii	$\frac{1}{2}I\omega^2 = mg(4a + 5a \sin \theta)$	M1	Equation involving KE and PE	
		ii	$\frac{50}{3}ma^2\omega^2 = mga(4 + 5 \sin \theta)$	A1	FT	
		ii	Angular speed $\omega = \sqrt{\frac{3g(4 + 5 \sin \theta)}{50a}}$	A1	AG <u>Examiner's Comments</u> In part (ii) the methods for finding the angular speed and angular acceleration were well understood and usually applied accurately.	
		iii				Or differentiating energy equation
		iii	$mg(5a \cos \theta) = I\alpha$	M1	Equation of rotational motion	Or writing $\omega \frac{d\omega}{d\theta}$ in terms of θ



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Angular acceleration is $\frac{3g \cos \theta}{20a}$	A1	Accept $\frac{15g \cos \theta}{100a}$ etc <u>Examiner's Comments</u> In part (iii) the methods for finding the angular speed and angular acceleration were well understood and usually applied accurately.	
		iv		M1	For radial acceleration $r \omega^2$	
		iv	$R - mg \sin \theta = m(5a)\omega^2$	A1		
		iv	$R - mg \sin \theta = \frac{3}{10}mg(4 + 5 \sin \theta)$			
		iv	$R = \frac{1}{10}mg(12 + 25 \sin \theta)$	A1		
		iv		M1	For transverse acceleration $r \alpha$	Or use of $I_G \alpha$
		iv	$mg \cos \theta - S = m(5a)\alpha$	A1		Or $S(5a) = (\frac{25}{3}ma^2)\alpha$
		iv	$mg \cos \theta - S = \frac{3}{4}mg \cos \theta$			
		iv	$S = \frac{1}{4}mg \cos \theta$	A1	<u>Examiner's Comments</u> In part (iv) there were many mistakes in forming the equations of motion to find R and S. These included sign errors, omission of the mass, omission of the radius (5a), omission of the weight term and confusion between sine and cosine. Sometimes the radial and transverse accelerations were interchanged.	
			Total	14		



Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	By energy	M1	Need correct 3 terms; allow	$v^2 = 2ag - 2ag \cos \theta$ Examiner's Comments This part proved straightforward for most candidates, though some made sign errors and some failed to include the component of the weight term when finding an expression for T . A small number of candidates left the answer as θ , or $\cos \theta$, failing to find the height required.
		i	$\frac{1}{2}m(4ag) = \frac{1}{2}mv^2 + mga(1 + \cos \theta)$ Use of $F = ma$	A1	wrong sign, missing/extra g , missing m / a ; sin for cos correct Need 3 terms and g , allow sign	
		i	$T + mg \cos \theta = \frac{mv^2}{a}$	M1	slip, sin for cos	
		i	$T = 2mg - 3mg \cos \theta$	A1		
		i	Slack when $\cos \theta = \frac{2}{3}$	M1	Ft if their T has right form	
		i	Height is $\frac{5}{3}a$	A1		
		ii	If $\theta > \pi/2$, $\frac{1}{2}mU^2 > mga$	M1	Allow '=' for all M marks	$\sqrt{2ag} < U < \sqrt{5ag}$ u is vel at top
		ii	$U > \sqrt{(2ag)}$	A1	Allow $\geq$	
		ii	For no complete revolutions $\frac{1}{2}mU^2 < \frac{1}{2}mu^2 + 2mga$	M1	Allow wrong sign	
		ii	and $mg = m \frac{u^2}{a}$	M1	Allow wrong sign	
		ii	$U < \sqrt{(5ag)}$	A1	Allow $\leq$	
		ii	OR Use $\frac{1}{2}mU^2 = \frac{1}{2}mv^2 + mga(1 + \cos \theta)$			
		ii	and $T + mg \cos \theta = \frac{mv^2}{a}$	(M1)		
		ii	To get $T = m \frac{u^2}{a} - 2mg - 3mg \cos \theta$ oc	(A1)		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	When $T = 0$, $U^2 = 2ag + 3ag \cos \theta$	(M1)	Allow $\leq, <$	Examiner's Comments Many candidates tackled this by finding the tension in terms of U for a general position, and then putting it equal to zero. However, the majority using this method failed to progress any further, and those that did almost always only found the greatest possible value of U for the string to not remain taut in the subsequent motion. Similarly, other candidates who looked at the specific position where the string went from being taut to not being taut (or vice versa) almost only considered the topmost point on the motion and ignored the fact that the string will always be taut if the angle it makes with the upward vertical is more than 90°.
		ii	$(\theta = 0)$ gives $U < \sqrt{5ag}$	(A1)	Allow $\leq$	
		ii	$(\theta = \frac{\pi}{2})$ gives $U > \sqrt{2ag}$	(A1)	Allow $\geq$	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	Conservation of energy	M1	Examiner's Comments	Need 4 terms; allow sign & trig errors
					Many candidates did this question completely correctly. a few made things more difficult than necessary for themselves by using a zero level for potential energy that was not the initial position, others made a sign error, or forgot about the kinetic energy of one of the particles. Quite a number of candidates made a sign error in the equation	
					$0.4g \sin \theta - R = \frac{0.4v^2}{a}$ and so did not find a correct expression for R ; a small number omitted the weight term. Almost all candidates knew that depended on putting $R = 0$, and those who had part (i) correct invariably gained both marks	
		i		M1		Both KE or both PE correct
		i	$\frac{1}{2} 0.4v^2 + \frac{1}{2} 0.6v^2 + 0.4ga \sin \theta - 0.6ga \theta = 0$	A1		completely correct
		i		M1	Attempt to find v^2 dep both earlier M1s	Allow with sign and trig errors
		i	$v^2 = 3.92a \quad (3 < 9 - 2\sin < 9)$	A1	AG	No errors
		i	$F = ma$ radially for P	M1*		Allow sign and trig errors
		i	$0.4g \sin \theta - R = \frac{0.4v^2}{a}$	A1		
		i		*M1	Manipulation attempted, leading to $a\theta + b \sin \theta$	Allow sign and trig errors
		i	$R = -4.7040 + 7.056 \sin \theta$	A1		2.352 $(-2 \theta + 3 \sin \theta)$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Using $R = 0$	M1	$0 = -4.704 (9 + 7.056 \sin \theta)$ Examiner's Comments Many candidates did this question completely correctly. A considerable number of candidates omitted this part.	
		ii	$(k =) \frac{2}{3}$	A1		Must be from correct expression in (i)
			Total	11		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	Using $F = ma$	M1	Must have the right 3 terms; allow sign error / sin for cos for M1	$v^2 = u^2 - 9.8(1 - \cos\theta)$
		i	$T - 0.2g\cos\theta = 0.2v^2/0.5$	A1		
		i	By energy	M1		
		i	$\frac{1}{2} \times 0.2u^2 = \frac{1}{2} \times 0.2v^2 + 0.2g \times 0.5(1 - \cos\theta)$	A1		
		i	$T = 5.88\cos\theta + 0.4u^2 - 3.92$	A1	AG with no errors and no gaps in argument	
					Examiner's Comments	
					Most candidates had little trouble with this part question, though some did not show sufficiently clearly how they derived the given answer. A small number of candidates, having initially failed to arrive at the given answer, did the question again, or went back and corrected their work.	
		ii	When $\theta = 180^\circ$, $5.88\cos\theta + 0.4u^2 - 3.92 = 0$	M1	Allow inequalities for M1A1	4.9497... Not > 4.95
		ii	$-5.88 + 0.4u^2 - 3.92 = 0$	A1		
		ii	Min u is 4.95 (m s ⁻¹)	A1	$\frac{7}{2}\sqrt{2}$	
		ii	OR, at top, $mg = \frac{mv^2}{r}$, so $v^2 = 0.5g$	B1	Allow inequalities for B1M1	
		ii	By energy $\frac{1}{2} 0.2u^2 = \frac{1}{2} 0.2 \times 0.5g + 0.2g$	M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Min u is $4.95 \text{ (m s}^{-1}\text{)}$	A1	Examiner's Comments Quite a number of candidates, including some with high marks, assumed that the particle would be stationary at the highest point, and arrived at the wrong answer of 4.43 m s^{-1} . Other candidates thought the value of θ at the uppermost point was $0, \pi/2$ or 2π	
		iii	$5.88\cos\theta + 0.4 \times 12.25 - 3.92 = 0$	M1		
		iii	$\cos\theta = (3.92 - 4.9)/5.88 (= -1/6)$	A1	Might see $\theta 99.6^\circ$ or 1.74 radians	$99.49406\dots^\circ, 1.73824\dots\text{rads}$
		iii	Use energy eq ⁿ from (i)	M1	Accept use of their θ	
		iii	$\frac{1}{2} \times 0.2 \times 3.5^2 = \frac{1}{2} \times 0.2 v^2 + 0.2g \times 0.5(1 - \cos\theta)$ $v = 0.904 \text{ m s}^{-1}$	A1	$\frac{7}{30}\sqrt{15}$	$0.903696\dots$
		iii	OR use T equation from (i)	M1		
		iii	$0 - 0.2g(-1/6) = 0.2v^2/0.5$ $v = 0.904 \text{ m s}^{-1}$	A1	Examiner's Comments The most common error on this part was to make a sign error and arrive at $\cos\theta = \frac{1}{6}$. A significant minority of candidates found the angle θ and then stopped before finding the speed.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	$\frac{1}{2}mv^2 = mg \times 0.8(1 - \sin 30^\circ)$	M1	Or with '5m' if for Q	Allow g missing for M1.
		i	$v = 2.8 \text{ m s}^{-1}$	A1		Might see $v^2 = 0.8g$
		i	Speed of P and Q equal	B1ft	soi	
		i	Use conservation of momentum			
		i	$5m \times 2.8 - m \times 2.8 = 5mq + mp$	B1ft	Ft on velocity	p is vel of P, q is vel of Q, both to left
		i	Use of NEL	M1		Allow $\pm e$
		i	$p - q = -0.95(-2.8 - 2.8)$	A1ft	Ft on velocity	
		i	$p = 6.3 \text{ m s}^{-1}$	A1	Supporting work required for AG	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$q = 0.98 \text{ m s}^{-1}$ Q moves to left	A1	Direction must be clear Examiner's Comments Most candidates realised that P and Q would collide at the lowest point of the cylinder, with equal speeds. However, a number of candidates, having realised that the particles would collide at the lowest point, correctly worked out the speed of P, but made an error in the speed of Q, usually through having the mass of Q as m in the expression for KE and $5m$ in the expression for PE. Some follow through marks for the rest of the part were available to these candidates. A small number of candidates worked with the angle that a line joining O to the point of collision makes with the vertical; these candidates generally made no worthwhile progress. A small number of candidates were completely correct, except for not making clear the direction of motion of Q after the collision.	
		ii	By energy for P at top	M1	Must have 3 terms	Allow g missing, sign error
		ii	$\frac{1}{2}m6.3^2 = \frac{1}{2}mv^2 + mg \times 1.6$	A1		
		ii	$v^2 = 8.33$	A1	Soi	
		ii	Use $F = ma$ at top	M1	Must have 3 terms	Allow g missing, sign error
		ii	$mg + R = m \times \frac{8.33}{0.8}$	A1ft	Their v^2	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$R = 0.6125m$	A1CAO	Or $49m/80$ Examiner's Comments This part was done very well. The most common errors were to use the wrong height from the bottom of the cylinder to D , and to miss out the weight term when working out the normal reaction acting on P at D . A small number of candidates worked this out in terms of a general angle, putting that angle equal to 180° , usually correctly, at the end.	
			Total	14		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	$mg \times 0.2 = \frac{2.45m \times e}{0.3}$	M1		Allow sin / cos, wrong sign, missing g
		i	$e = 0.24$	A1	No errors; must show all numbers Examiner's Comments Most candidates did this completely correctly, though not all candidates showed all the values needed to calculate the extension.	
		ii	Use $F = ma$ down slope	M1	3 terms needed	Allow sign error, sin / cos, missing g or m
		ii	$mg \sin \alpha - \frac{2.45m(x - 0.3)}{0.3} = m\ddot{x}$			Could use x in place of $x - 0.3$, leading to $\ddot{x} = -\frac{49}{6}$
		ii	$\ddot{x} = -\frac{49}{6}(x - 0.54)$	A1	oe Accept 2.45/0.3 for to ω^2	$(x - 0.24)$ (about $x = 0.24$) Or $x + 0.24$ in place of $x - 0.3$ leading to $\ddot{x} = -\frac{49}{6}x$ (about $x = 0$)
		ii	SHM (about $x = 0.54$)	A1	Dep M1A1. Must be in correct form, and ω^2 in simplified form	
		ii	$\omega = 7/\sqrt{6}$ (2.8577)	B1	Soi	May see $\omega^2 = 8\frac{1}{6}$
		ii	$T = 2.20$	B1CAO	AG Need to see $2\pi/\omega$ oe	2.1986568...

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$a = 0.105 \text{ m (0.1049795)}$	B1ft	ft their $\omega = \frac{3\sqrt{6}}{70}$ Examiner's Comments Most candidates successfully established SHM, usually using $(x + 0.24)$ as the extension of the string, following on from the calculation in part (i). In a small number of cases there were errors or slips in the working, though most candidates ended up with the right value for 'ω^2', and were able to access the marks for establishing the period and amplitude. Some candidates lost a mark by failing to point out that an equation of the form $\ddot{x} = -\omega^2 x$ represents SHM. Some candidates worked with energy, which if done for a general point followed by differentiation with respect to x can establish SHM, though very few correct solutions of this form were seen. Most candidates who used energy used it to find the amplitude of the motion, and were then able to access the marks for showing that the period was 2.20s.	NB Can find a by energy, leading to ω and T
		iii	Use of SHM eqn for distance	M1	$x = a \sin \omega t$	Allow M1 for $x = a \cos \omega t$
		iii	$x = -0.0956(227\dots)$	A1ft	Their a	Or -0.9553 or -0.09577
		iii	Dist from O is $0.444(377\dots)$ (m)	A1CAO		
		iii	Use of SHM equation for velocity	M1	$v = a \omega \cos \omega t$	Allow M1 for $v = -a \omega \sin \omega t$ if consistent with x eqn for $\sin / \cos, a, \omega$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$v = -0.124$ ($-0.123949\dots$)	A1	Must be clear velocity is towards O Examiner's Comments This was done well by candidates who had been successful in part (ii). Some follow through marks were allowed for candidates who had calculated the amplitude wrongly. Not all candidates went on after finding the displacement at time 1.5s to find the distance of P from O. It should be pointed out that candidates using $v^2 = \omega^2 (a^2 - x^2)$ to find the speed of P needed to establish the direction of motion if they were to gain the final mark.	Use of $v^2 = \omega^2 (a^2 - x^2)$ will not gain A1 unless direction is established
			Total	13		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i	Use of energy equation at A and B	M1	3 terms needed $mg0.6\cos\frac{\pi}{6} = mg0.6\cos\theta + \frac{1}{2}mv^2$ Examiner's Comments This proved to be very straightforward for a large number of candidates who had been well trained and knew exactly what to do. Since the answer was given as a surd the final mark was not awarded to those who expressed $\sin(\pi/6)$ as a decimal in their working. However, quite a number of candidates made errors; some got the PE at A and B the other way round, and so had a negative quantity for v^2; others made a sign error in using $F = ma$; and some candidates had the particle start at T so that $\pi/6$ did not appear in the energy equation.	allow sign error, missing $m / g / r$
		i		A1		allow if θ replaced by $\phi + \pi / 6$
		i	$F = ma$ radially	M1		allow sign error, missing m / g
		i		A1	$mg \cos \theta - R = \frac{mv^2}{0.6}$	
		i	Use of $R = 0$	M1	May be incorporated in previous step	
		i	$\cos TOB = \frac{\sqrt{3}}{3}$ AG	A1	Completely correct	not given if decimals used for angle.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Use of $\sqrt{3}/3$ in 'correct' equation in (i)	M1	$mg0.6\cos\frac{\pi}{6} = mg0.6 \times \frac{\sqrt{3}}{3} + \frac{1}{2}mv^2$ or $mg\frac{\sqrt{3}}{3} = \frac{mv^2}{0.6}$ Examiner's Comments Candidates who did part (i) correctly usually got this part right. Some who made errors in part (i), or made little attempt at part (i), also got this right by showing full correct working here.	equation must have gained M1 in (i) but allow restart here
		ii	1.84 (ms ⁻¹)	A1		
		iii	Use of $F = ma$ tangentially	M1	$mg\sin\theta = ma$ seen Examiner's Comments This proved a difficulty to some candidates, with some working out $g \cos \theta$, others not using an angle, and others not being able to work out $\sin \theta$ correctly.	allow missing m / g , – sign; allow M1 if angular accel found
		iii	8.00 (ms ⁻²)	A1		
			Total	10		

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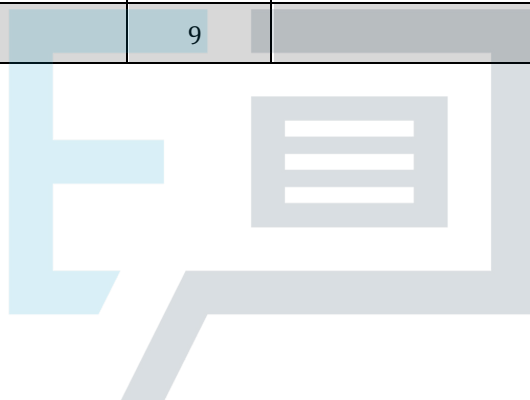
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Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i		M1*	Resolving vertically (3 terms)	
		i	$T \cos 30 + R \sin 60 = mg$	A1		
		i		M1*	Resolving horizontally (3 terms); an r used where r is not just a	
		i	$T \sin 30 - R \cos 60 = m(a \sin 30) \omega^2$	A1		
		i		M1 dep*	Eliminating T and solve for R in terms of m, g, a and ω	
		i	$R = \frac{1}{6}m(2\sqrt{3}g - 3a\omega^2)$	A1	AG Correctly shown	
					Examiner's Comments	
					The usual method of resolving parallel to and perpendicular to the direction of the acceleration was successfully employed by the majority of candidates. However some used the wrong radius in their acceleration and others found the algebraic manipulation to achieve the given answer a challenge. A minority of candidates attempted to resolve in other directions with not much success, failing to realise that the acceleration has a component in the direction they were resolving.	
		ii	For using $R = 0$ to attempt to find either v or T	M1	Or attempt to find ω	
		ii	$T = \frac{mg}{\cos 30} = 39.6$	A1	39.606228...	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\omega^2 = \frac{T}{ma}$, $v = 1.19 \text{ ms}^{-1}$	A1	1.1893309... Examiner's Comments Most candidates were aware that using $R = 0$ was the way to proceed in this question. However two requests were made with a significant number not attempting to find the tension in the string. In other cases, candidates only attempted the angular speed despite the request being for the speed of the particle.	
			Total	9		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
29			$T\cos\theta = 0.3g$ (= 2.94)	B1	Resolve vertically or $T\sin\alpha = 0.3g$; component of T required	
				M1	Attempt at use of N2L with $a = r\omega^2$ or $a = v^2/r$	
			$T\sin\theta = 0.3(0.343)(6^2)$ (= 3.7044)	A1	Resolve horizontally or $T\cos\alpha = 0.3(0.343)(6^2)$; component of T required	
				M1	Attempt to solve for T or θ ; component of T required in both equations	
			$T = 4.73 \text{ N}$	A1		
			$\theta = 51.6^\circ$	A1	SC if 38.4° then B1M1A1M1A1A0 5/6 Max	
					Examiner's Comments	
					This question proved a good source of marks for many candidates. The only significant error that was made was for candidates to use the angle with the horizontal and not all candidates went on to give the angle with the vertical as requested.	
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i		M1	Attempt at use of N2L with $a = v^2/r$ or $a = r\omega^2$; allow verification.	
		i	$0.4v^2/0.6 = 6, v = 3$	A1	AG If $+/-3$ then -3 must be rejected for A1. Examiner's Comments Not all candidates recognised that since the answer was given there was a need for them to show their working clearly. There were many good solutions but a disappointing number wrote a formula and then gave a value for ω or v^2 without showing the values used to obtain it. A few candidates did not know how to proceed and seemed to think the coefficient of restitution was relevant in this part.	
		ii		*M1	Attempt at use of conservation of linear momentum	
		ii	$3(0.4) = 0.4v_A + 0.5v_B$	A1		
		ii		*M1	Attempt at use of restitution equation, must be correct way round	
		ii	$v_B - v_A = 3(0.35)$	A1	Must be consistent with the directions used for CoLM	
		ii		dep*M1	Solving simultaneous equations	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v_A = 0.75 \text{ m s}^{-1}$ and $v_B = 1.8 \text{ m s}^{-1}$	A1	Both values positive as final answer. Examiner's Comments The majority of candidates scored full marks on this part. For those who guessed the wrong direction for one of their velocities, the issue was resolved at the end and speed was stated successfully.	
		iii	$Y = 0.4g$ (= 3.92)	B1	Resolve vertically	
		iii	$(4.9)^2 - (0.4g)^2$ (= 2.94 ²)	B1	Use of Pythagoras	
		iii	$0.4(0.6)\omega^2 = cv(2.94)$	M1	Resolve horizontally, $cv(2.94)$ from a legitimate attempt at 2.94	
		iii	$\omega = 3.5$	A1	Exact	
		iii	OR $4.9\cos\theta = 0.4g$	B1	Resolve vertically, θ angle with vertical	
		iii	$\cos\theta = 0.8$ or $\theta = 36.9$	B1		
		iii	$0.4(0.6)\omega^2 = 4.9\sin\theta$	M1	Resolve horizontally, θ substituted and from a legitimate attempt	
		iii	$\omega = 3.5$	A1	Exact	
		iii	OR $Y = 0.4g$ (= 3.92)	B1	Resolve vertically	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$X = 0.4(0.6)\omega^2 (= 0.24\omega^2)$	B1	Resolve horizontally	
		iii	$(0.4g)^2 + (0.4(0.6)\omega^2)^2 = 4.9^2$	M1	Attempt at use of Pythagoras, from a legitimate attempt at X	
		iii	$\omega = 3.5$	A1	Exact	
					Examiner's Comments	
					This part was poorly done by a significant majority of candidates, who were unable to find the force required to proceed. Many thought that 4.9 N was the force towards the centre. Other common errors seen were for the force towards the centre to $(4.9 - 0.4g)$ N, as well as some who thought the 4.9 N force was at an angle of 45° to the horizontal.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i		M1	Resolve vertically (3 terms); may be different T 's at this stage	
		i	$T\cos 30 + T\cos 45 = 0.4g$	A1		
		i	$T = 2.49 \text{ N}$	A1	$T = 2.4918\dots$ Examiner's Comments This was well answered. The majority appreciating that the tensions would be equal.	
		ii		M1	Resolve horizontally (3 terms); may be different T 's at this stage	
		ii	$cv(T)\sin 30 + cv(T)\sin 45 = 0.4v^2/0.5$	A1	Or use acceleration $= 0.5\omega^2$	
		ii	$v = 1.94 \text{ m s}^{-1}$	A1	$v = 1.93904\dots$ Examiner's Comments Well answered by the majority. Only a small number of candidates found angular speed and then didn't convert to speed as requested.	
	Copyright © 2024 Exam Papers Practice	iii	$(2AP =) \frac{0.5}{\sin 45} + \frac{0.5}{\sin 30}$	M1	Reasonable attempt to use trigonometry to find total length of string	
		iii	$AP = 0.854 \text{ m}$	A1	AG ($AP = 0.85355\dots\text{m}$) Examiner's Comments This part was well done by those who realised that trigonometry was to be used. Some mistakenly thought that circular motion formulae needed to be used.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$2T\sin\theta = 0.4(0.854\sin\theta)(3.46^2)$	M1	θ angle with vertical. Resolve horizontally. Allow with T only. r = component of 0.854	
		iv	$T = 2.04 \text{ N}$	A1	$T = 2.04474... \text{ N}$ using AP = 0.854 m, $T = 2.04367... \text{ N}$ using exact AP	
		iv	$2T\cos\theta = 0.4g$	M1	θ angle with vertical. Resolve vertically. Allow with T only	
		iv	$\theta = 16.5^\circ$ or 16.6°	A1	$\theta = 16.55377...^\circ$ using AP = 0.854 m, $\theta = 16.4526...^\circ$ using exact AP SC M1A0M1A1 for use of T instead of 2T throughout	
					Examiner's Comments	
					The main difficulty in this part was the failure to recognise that the radius of the circular motion was no longer 0.5 m. It was common also to see only one tension used although these candidates still had access to the majority of the marks.	
			Total	12		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
32		i	Vertical force = mg	*B1		
		i	Horizontal force = $m \times 0.4 \times 7^2$	*M1A1		
		i	Uses vertical force = $\mu \times$ horizontal force	dep*M1	Dependent on B1 and M1	
		i	$\mu = 0.5$	A1	If a value for m used B0M1A0M1A0 max. Examiner's Comments It is surprising how many used the string and omitted it.	
		ii	$mg = T \times 0.3/0.5$	B1		
		ii		*M1	Resolve T and equate to mass $\times (r\omega^2$ or $v^2/r)$	
		ii	$m \times 0.4 \omega^2 = T \times 0.4/0.5$	A1		
		ii	Solve for ω or v	dep*M1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\omega = 5.72 \text{ rad s}^{-1}$	A1	Allow $7\sqrt{6}/3$ If a value for m and/or T used B0M1A0M1A0 max. Examiner's Comments It is surprising how many used the string. Nevertheless, there were many correct solutions seen to both parts of the question. Those who were unsuccessful frequently presented confused diagrams. Forces labelled F often appeared in a number of different directions within the same diagram.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i	(a) $0.8F + 0.6R = 0.4g$	M1	Attempt to resolve vertically	
		i	$4F + 3R = 19.6$ [AG]	A1	www	
					Examiner's Comments	
					Some good solutions were seen. Some candidates either got a wrong angle or got a correct angle and resolved incorrectly, but amazingly still managed to get the correct result.	
		i		M1	Attempt with three terms.	
		i	(b) $0.8R - 0.6F = 0.4 \times 4.5^2/3$	A1	aef including cos, sin correct angle	
		i	Solve for R or F	M1	Use 2 relevant resolutions.	
		i	$F = 1.516$	A1		
		i	$R = 4.512$	A1		
		i	Use $\mu = F/R$ to get $\mu = 0.336$ [AG]	B1	Examiner's Comments	
					Finding a second appropriate equation proved difficult for quite a few. A remarkable number of combinations of values of F and R appear to produce $\mu = 0.336$! Some candidates lost marks here by eliminating both R and F from their equations to find just μ and so failing to find the values of R and F as required.	
		ii		M1	Resolve vertically, three terms	
		ii	$0.6R - 0.8F = 0.4g$	A1		
		ii	$R = 11.8$ or $F = 3.98$	A1		
		ii		M1	N2L, resolve horizontally, three	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$0.8R + 0.6F = 0.4 \times 3 \omega^2$	A1	terms	
		ii	$\omega = 3.14 \text{ rad s}^{-1}$	A1	Examiner's Comments This proved to be the most difficult question on the paper with very few correct solutions seen by examiners. Only a minority realised the direction of F had to change in this situation and most simply used the same equations as before, or used one equation with the values of R and F found in (ii), or solved equations in which F (or R) was zero.	
			Total	14		

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